CHAPTER 2: BASIC DIFFERENTIATION

In this chapter, we will introduce the concept of a derivative, covering the following topics:

- Introduction
- Conceptual Derivative
- Definition of a Derivative
- Power Rule
- List of Common Derivatives
- Product Rule
- Quotient Rule
- Chain Rule

Introduction: What is a Derivative?

Before we explain what a derivative is, it might be helpful to review the difference between secant lines and tangent lines.

- Secant line: A line that passes through *two points* on a line.
- Tangent line: A line that passes through *one point* on a line.
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To foreshadow, derivatives will "blur the boundaries" between these two lines.

Let's say that we have a function f continuous on $[a, b]$. What is the average rate of change over the interval?

We know that we can calculate the the *average rate of changeI, or slope* ($\frac{rise}{run}$) of a function over two points (x_1, y_1), (x_2, y_2).

The slope, m, between points (x_1, y_1) and (x_2, y_2) is given as follows:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

So, if we were asked to find the *slope* on the interval $[a, b]$, we would simply calculate the slope using points $(a, f(a))$, $(b, f(b))$.

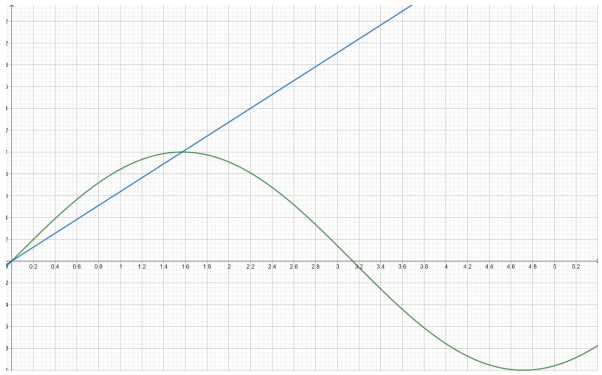
The slope, m, between points $(a, f(a))$, $(b, f(b))$ is given as follows:

$$m = \frac{f(b) - f(a)}{b - a}$$

For example, say that we were asked to compute the average rate of change for $sin(x)$ on the interval $[0, \frac{\pi}{2}]$. Using the slope formula, we see that

$$m = \frac{sin(\frac{\pi}{2}) - f(0)}{\frac{\pi}{2}} = \frac{1}{\frac{\pi}{2}} = \frac{2}{\pi}$$

So the average rate of change $sin\ (x)$ on the interval $[0,\ \frac{\pi}{2}]$ is $\frac{2}{\pi}$. That is, for every unit x increases/decreases, $f\ (x)$ increases/decreases by $\frac{2}{\pi}$ units.



Now, what if we were asked how fast the function was changing at *a point* rather than over *an interval*?

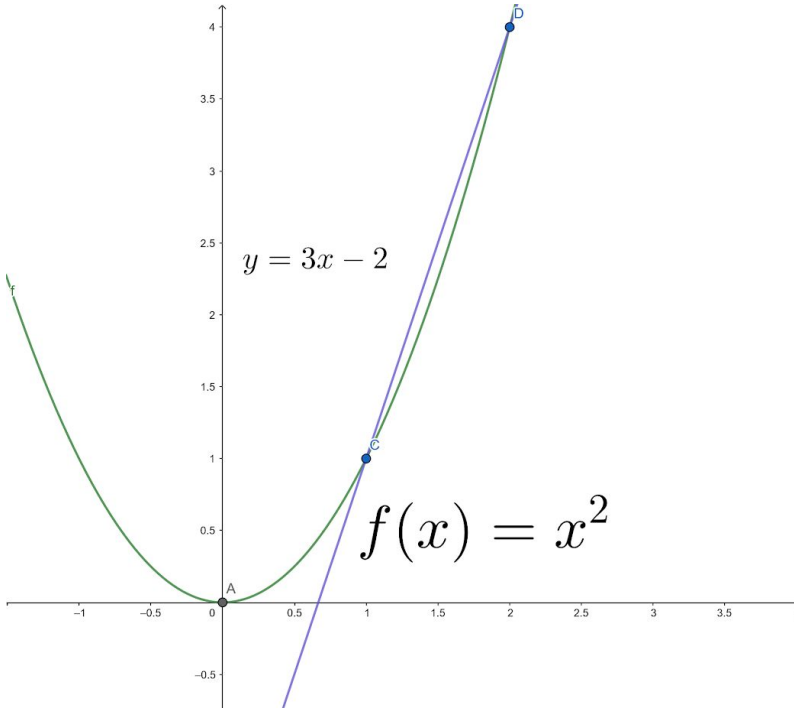
The answer to that is one of the fundamental questions in single variable calculus.

Let's take a look at the following problem:

1. Find the slope of $f\ (x)\ =\ x^2$ at $x\ =1$.

How could we approach this problem?

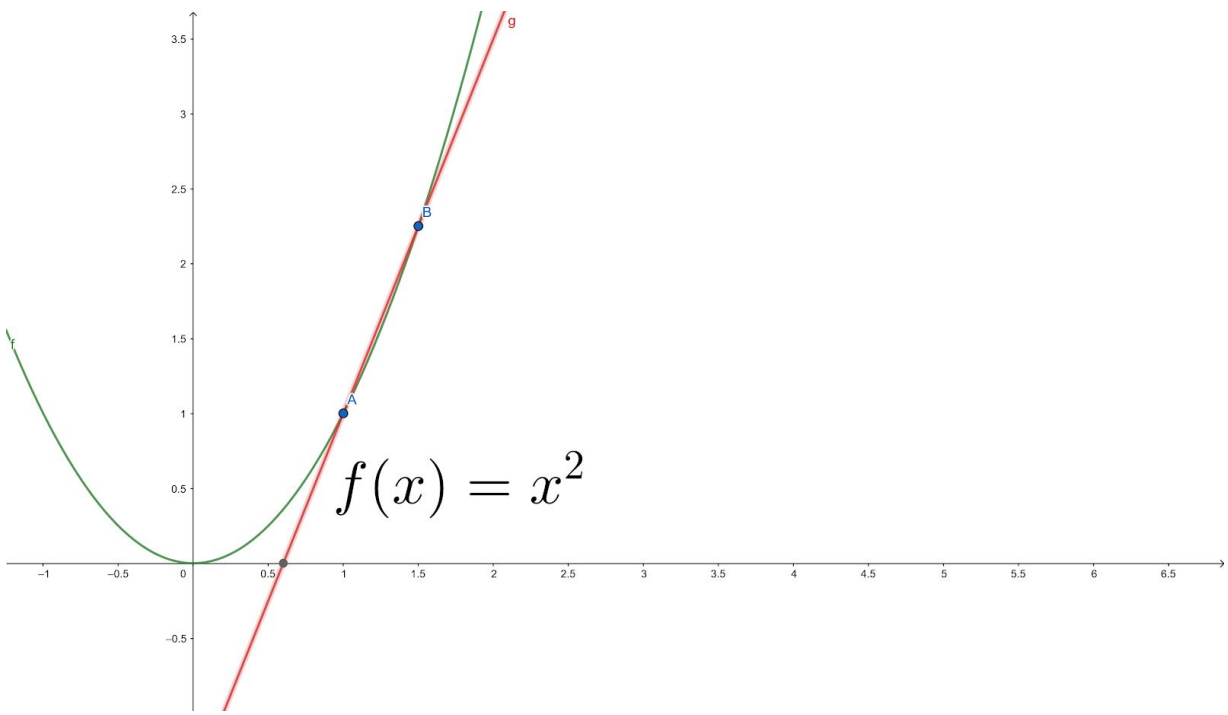
Well, we could start out by calculating the average rate of change (essentially the linear slope) between $(1,\ f\ (1))$ and another point $(b,\ f\ (b)$ on the graph. For the sake of convenience, let's use $(2,\ f\ (2))$.



Using the slope formula, we can easily see that the slope of this line is $\frac{f(b) - f(1)}{b - 1} = 3$ But we don't care about *this* slope: we want to know the slope *at* $x = 1$. So, what can we do?

Well, what if we were to shrink the distance between $(1, f(1))$ **and the other point,** $(b, f(b))$**?**

If we use, for example, $(1.5, f(1.5))$, we get that the average slope between $1, f(1))$ **and** $(1.5, f(1.5))$ is $\frac{f(1.5) - f(1)}{1.5 - 1} = \frac{2.25 - 1}{0.5} = 2.5$.



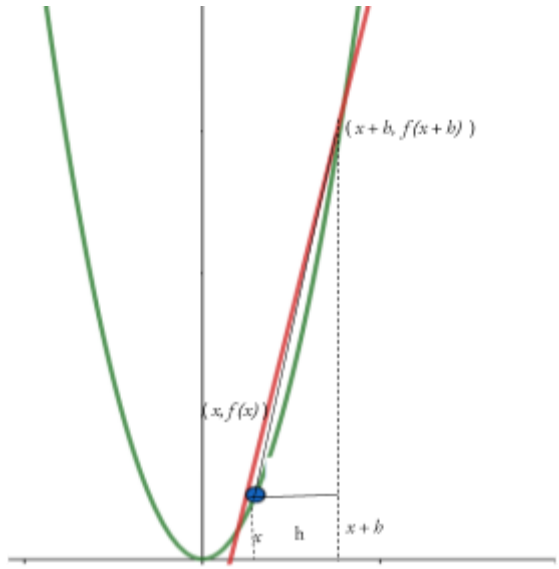
Slightly more accurate... but not close enough. Again, remember that we don't care about the slope between 2 points - we want to know the slope *at* a point.

In other words, we want to find the slope of the line *tangent* to $f(x)$ at $x = 1$. what this means is we want the line that touches $f(x)$ at precisely *1 point*. Notice above that the lines we used so far touches $f(x)$ at *2 points* (this is called a *secant line)*.

But how can we do this if we need 2 points to find the slope?

Recall the definition of a limit: As x approaches some value $x = a$, $f(x)$ approaches a corresponding $f(a)$, assuming the limit exists; that is, $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x)$.

Given a point $(a, f(a))$, we can rewrite $(b, f(b))$ as $(a + h, f(a + h))$, where $h = b - a$ (Think of h as the difference between points $x = a$ and $x = b$, or graphically, the distance between said 2 points as shown in the diagram below).



Now we can write the slope between $(a, f(a))$ and $(b, f(b))$ as follows:

$$m = \frac{f(a+h) - f(a)}{(a+h) - a} = \frac{f(a+h) - f(a)}{h}$$

Now what would happen if h approaches 0? We are essentially *shrinking* the distance between $(a, f(a))$ and $(b, f(b))$ until we get the infestinimal change in $f(x)$ with respect to x.

So to find the *instantaneous slope* at $x = a$, we take the limit as h approaches 0 of the slope of the secant line $\frac{f(a+h) - f(a)}{(a+h) - a}$.

We will find that

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{(a+h) - a} = f'(a)$$, where $f'(a)$ is the *derivative* of f at $x = a$.

This is the essence of derivatives: they describe the slope of a function *at one point.* If we were to think of slope as $\frac{y}{x}$, it would be represented as $\frac{a\ very,\ very\ small\ y}{a\ very,\ very\ small\ x}$, which is notated as $\frac{dy}{dx}$ where dy is an infinitesimal change in y and dx is an infinitesimal change in x.

In the above definition, we set h to be the infinitesimal change in x, or dx. To generalize for all x, the definition of the derivative can be written as follows:

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = f'(x)$$, where $f'(x)$ is the *derivative* of f for all x.

So back to the original problem: *Find the slope of* $f(x) = x^2$ *at* $x = 1$.

Using the definition of the derivative, we set $a = 1$ and take the limit as h approaches 0.

$$\lim_{h\to 0} \frac{(1+h)^2 - (1)^2}{h} = \lim_{h\to 0} \frac{1+2h+h^2 - 1}{h} = \lim_{h\to 0} \frac{2h+h^2}{h} = \lim_{h\to 0} 2 + h = 2 .$$

So, the slope of the tangent line to $f(x) = x^2 \; at \; x = 1$ is 2.

<u>Conceptual Derivative</u>

As mentioned earlier, the derivative of a function measures the *rate of change*, or the slope at set given points on the function.

Now in the previous example, we used the definition of the derivative to calculate the slope of a function at one specific point. However, we can use the same process to determine the *derivative function*; that is, for a function f, the derivative function f' outputs the slope of any point x on f.

What is the derivative of $f(x) = x^2$?

Knowing the definition of the derivative $\lim_{h\to 0} \frac{f(x+h) - f(x)}{(x+h) - x}$, we can solve as follows:

$$\lim_{h\to 0} \frac{(x+h)^2 - (x)^2}{h} = \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} = \lim_{h\to 0} 2x + h = 2x$$

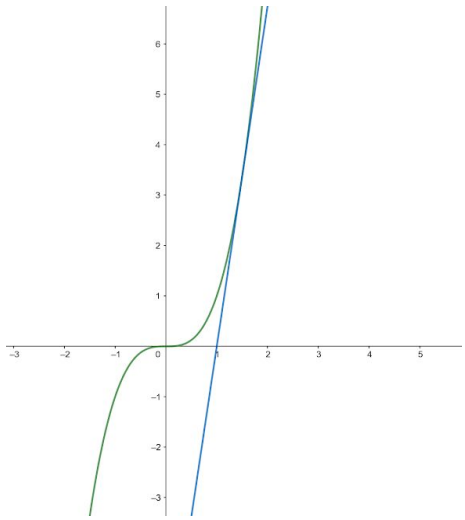
What does this mean? Well, for any point $x = a$ on the domain of $f(x) = x^2$ (which in this case conveniently happens to be all real numbers), the slope of the line tangent to that point is $2a$.

For instance, at $x = 2$, the slope of the line tangent to the graph at that point is $2(2) = 4$. At $x = -5$, the slope is $2(-5) = -10$. At $x = 0$, the slope of *that* tangent line is $2(0) = 0$; in other words, the line tangent to f at $x = 0$ is horizontal.

What about the derivative of $f(x) = x^3$?
Let's use the definition of the derivative again:

$$\lim_{h\to 0} \frac{(x+h)^3 - (x)^3}{h} = \lim_{h\to 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} = \lim_{h\to 0} 3x^2 + 3xh + h^2 = 3x^2$$

So, for the function $f(x) = x^3$, the slope of the tangent tangent to any point a on the function is simply $3a^2$. For instance, in the diagram below, the line tangent to $x = 1.5$ has a slope of $3(1.5)^2 = 6.75$.



What about the derivative of $f(x) = sin(x)$? This one is trickier since f isn't a polynomial function, but it's still doable: We need to rely on the trigonometric identity $sin(\alpha + \beta) = sin(\alpha)cos(\beta) + cos(\alpha)sin\beta)$:

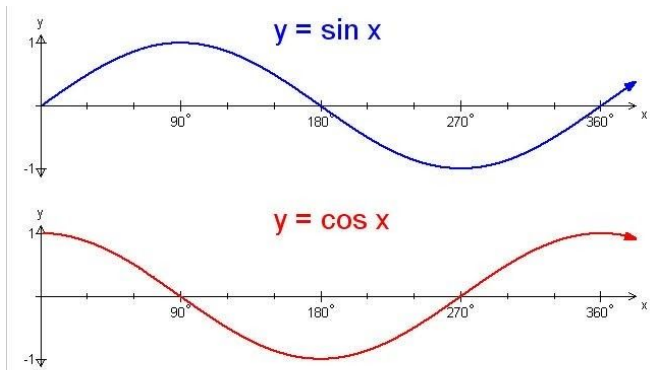
$$\lim_{h \to 0} \frac{sin(x+h) - sin(x)}{h}$$
$$= \lim_{h \to 0} \frac{sin(x)cos(h) + sin(h)cos(x) - sin(x)}{h}$$
$$= \lim_{h \to 0} \frac{sin(x)(cos(h) - 1) + sin(h)cos(x)}{h}$$
$$= \lim_{h \to 0} \frac{sin(x)(cos(h) - 1)}{h} + \frac{sin(h)cos(x)}{h}$$
$$= sin(x) \lim_{h \to 0} \frac{(cos(h) - 1)}{h} + cos(x) \lim_{h \to 0} \frac{sin(h)}{h} *$$
$$= cos(x)$$

Turns out the derivative function for $f(x) = sin(x)$ is just $cos(x)$!

*To solve these limits, we can use either the Squeeze Theorem or a powerful technique for solving indeterminate limits called **L'Hôpital's Rule**. Don't worry about either for now; for AP Calculus ®, all you have to do is memorize the derivatives of basic functions.

Again, what this means is that at any point a on the function $f(x) = sin(x)$, the instantaneous slope at that point is equal to $cos(a)$.

Confused? Let's take a look at the graphs of $sin(x)$ and $cos(x)$ more closely:



Notice how on the interval $(0, \frac{\pi}{2})$, the graph of $sin\ (x)$ is increasing, signifying a positive, albeit non constant (changing) slope. At $x = \frac{\pi}{2}$, the slope of the line tangent to $sin\ (x)$ is zero; in other words, the line is horizontal. On the interval $(\frac{\pi}{2}, \pi)$ the graph of $sin\ (x)$ is decreasing, signifying a consistently negative slope.

Now look at the behavior of $cos\ (x)$. On the interval $(0, \frac{\pi}{2})$, the *function* (*not the slope*) is positive (above the x-axis). At $x = \frac{\pi}{2}$, the *function* intersects the x-axis/is equal to 0. On the interval $(\frac{\pi}{2}, \pi)$, the *function* (*not the slope*) is negative (below the x-axis).

What does this mean?

As the *derivative* of $sin\ (x)$, the graph of $cos\ (x)$ essentially maps out the slope of $sin\ (x)$: Whenever the *slope* of $sin\ (x)$ is positive/negative, the *graph* (output values) of $cos\ (x)$ will be positive/negative as well. We will elaborate more on relating a function with its derivatives in Chapter 4; for now, all you really need to understand is that the *derivative function is a measure of the rate of change of the original function.*

<u>A Quick Note: Derivative Notation:</u>

Leibniz notation denotes the infinitesimal change in y with respect to x as $\frac{dy}{dx}$. For the infinitesimal change in a function f with respect to x, we denote this as $\frac{df}{dx}$.
We can think of the expression $\frac{d}{dx}$ itself as an operator. For instance,
$\frac{d}{dx}x^3 = 3x^2$.

Lagrange (also called Newton's notation, but Newton stole it from Lagrange) Notation denotes the derivative of a function f with respect to x as $f'(x)$ ("*f prime of x*"). The 2nd derivative is written as $f''(x)$, and all higher order n^{th} derivatives are written as $f^{(n)}(x)$.
For instance, if $f\ (x) = x^3$, $f'\ (x) = 3x^2$

<u>Definition of A Derivative</u>

The *derivative* (instantaneous slope) of any differentiable real-valued function at $x = a$ can be written as either the following forms:

$$\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = f'(x)$$

$$\lim_{x\to a} \frac{f(x)-f(a)}{x-a} = f'(a)$$

If $f'(x)$ exists, then f has a derivative or is differentiable at that point x.

<u>Differentiability</u>

A function f is differentiable at c (that is, $\lim_{h\to 0} \frac{f(c+h)-f(c)}{c}$ exists) if:

1. $\lim_{x\to c} f(x) = f(c)$

The function must first be continuous at $x = c$ for it to be differentiable, although the converse of this statement is not necessarily true. A function can be continuous at some value c but not differentiable at c, or even continuous everywhere but differentiable nowhere (although you will never come across such a function in AP Calculus ®).

2. $\lim_{x\to c^-} f'(c) = \lim_{x\to c^+} f'(c)$ (the slope of the derivative is the same left and right of the point)

Additionally, if, $\lim_{h\to 0} \frac{f(c+h)-f(c)}{h} = \pm\infty$, the tangent line at c is vertical and hence non-differentiable. Likewise, if $\lim_{h\to 0^-} \frac{f(c+h)-f(c)}{h} = \pm\infty$ and $\lim_{h\to 0^+} \frac{f(c+h)-f(c)}{h} = \mp\infty$, there exists a vertical cusp at c, and f is hence not differentiable at that point.

To summarize, a function is not differentiable at $x = c$ in the following scenarios:

Scenario:	Example:
1. There exists a **corner** at $x = c$	$f(x) = \lvert x\rvert$ at $x = 0$
2. There exists a **cusp** at $x = c$	$f(x) = x^{\frac{2}{3}}$ at $x = 0$
3. There exists a **vertical tangent** at $x = c$	$f(x) = x^{\frac{1}{3}}$ at $x = 0$

<u>The Power Rule</u>

We now know we can differentiate functions directly using the definition of a derivative.

<u>Exercise 2.1:</u>

Differentiate the following functions using the definition of the derivative. The solutions are right below on this page, but we recommend attempting the problems first before viewing the answers.

1. $f(x) = 3x^2$
2. $f(x) = x^4$
3. $f(x) = 7x$

<u>Solutions:</u>

1. $\lim_{h\to 0} \frac{3(x+h)^2 - 3(x)^2}{h} = \lim_{h\to 0} \frac{3x^2+6xh+3h^2 - 3x^2}{h} = \lim_{h\to 0} 6x + 3h = 6x$
2. $\lim_{h\to 0} \frac{(x+h)^4 - x^4}{h} = \lim_{h\to 0} \frac{x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4 - x^4}{h} = \lim_{h\to 0} 4x^3 + 6x^2h + 4xh^2 + h^3 = 4x^3$
3. $\lim_{h\to 0} \frac{7(x+h) - 7x}{h} = \lim_{h\to 0} \frac{7(x+h) - 7x}{h} = \lim_{h\to 0} \frac{7x + 7h - 7x}{h} = \lim_{h\to 0} 7 = 7$

Do you notice a pattern yet?

We already saw earlier that $\frac{d}{dx}x^3 = 3x^2$ and $\frac{d}{dx}x^2 = 2x$. Now we see that:

- $\frac{d}{dx}x^4 = 4x^3$
- $\frac{d}{dx}3x^2 = 6x$
- $\frac{d}{dx}7x = 7$

If there is a pattern or algorithm of some sort, it would probably be a *lot* easier to apply that algorithm to compute derivatives more quickly…

Introducing **THE POWER RULE:**

For a differentiable function of the form $f(x) = x^n$, $\frac{df}{dx} = nx^{n-1}$, $n \in \mathbb{R}$

We'll leave the proof as an exercise to the reader (which, by the way, is beyond the scope of AP Calculus ®), but it is crucial that you are able to remember the Power Rule (or else you better be able to derive it on the spot).

What does this mean? Well, if we were asked to find the derivative of x^5 , instead of using the definition of the derivative, we immediately know that $\frac{d}{dx}x^5 = 5x^4$ because of the Power Rule.

Much easier, right?

Notice that in the example problems above, we could have employed the Power Rule to compute the derivatives of the given functions rather than rely on the definition of the derivative.

- $\frac{d}{dx}x^4 = 4x^{4-1} = 4x^3$ (this should be pretty self-explanatory)
- $\frac{d}{dx}3x^2 = (2 \cdot 3)x^{2-1} = 6x$
- $\frac{d}{dx}7x = (1 \cdot 7)x^{1-1} = 7$

<u>Exercise 2.2:</u>
Differentiate the following functions. You can check your answer at the end of this unit:

1. $\frac{d}{dx}x^7$

2. $\frac{d}{dx}3x^4$

3. $\frac{d}{dx}x^5 + 3x^2 + 7x + 9$

4. $\frac{d}{dx}\frac{1}{x}$ **(Hint: Rewrite as an exponent)**

5. $\frac{d}{dx}5x^2 - \frac{1}{x^2}$

6. $\frac{d}{dx}\sqrt{x}$

7. $\frac{d}{dx}\frac{1}{\sqrt[3]{x}}$

<u>List of Common Derivatives:</u>

The Power Rule effectively gives us a quick way to compute the derivatives of any polynomial/rational function.
Unfortunately, there are other functions whose derivative cannot be directly computed using the Power Rule.

Before we go any further, it might be helpful if we had a list of the derivatives of common functions (instead of having to prove them all using the definition of the derivative, which is harder than it sounds).

So without further ado:

<u>Derivative of a Constant</u>
$\frac{d}{dx} C = 0$, where C is a constant

<u>Derivative of Absolute Value Function</u>

$\frac{d}{dx} |x| = \frac{|x|}{x} = \frac{x}{|x|}$ for all x except $x = 0$

<u>Derivatives of Trigonometric Functions</u>
$\frac{d}{dx} sin\,(x) = cos\,(x)$
$\frac{d}{dx} cos\,(x) = -sin\,(x)$
$\frac{d}{dx} tan\,(x) = sec^2\,(x)$
$\frac{d}{dx} sec\,(x) = sec\,(x)\,tan\,(x)$
$\frac{d}{dx} csc\,(x) = -cot\,(x)\,csc\,(x)$
$\frac{d}{dx} cot\,(x) = -csc^2\,(x)$

<u>Derivatives of Inverse Trigonometric Functions</u>
$\frac{d}{dx} sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}}$
$\frac{d}{dx} cos^{-1}(x) = -\frac{1}{\sqrt{1-x^2}}$
$\frac{d}{dx} tan(x) = \frac{1}{1+x^2}$

<u>Derivatives of Exponential and Logarithmic Functions</u>
$\frac{d}{dx} e^x = e^x$
$\frac{d}{dx} a^x = e^x ln\,(a)$, where $a \in \mathbb{R}$
$\frac{d}{dx} ln\,(x) = \frac{1}{x}$
$\frac{d}{dx} log_{\,b}\,(x) = \frac{1}{x\,ln\,(b)}$ where $b > 0$

It's probably a good idea to memorize the above derivatives, as it will most certainly be tested either directly or indirectly on the AP Calculus ® exam.

<u>The Product Rule:</u>

For differentiable functions $f(x)$ and $g(x)$, what is the derivative of $f(x)g(x)$? A common mistake that many students make is to presume that the derivative of $f(x) \cdot g(x) = f'(x) \cdot g'(x)$. Unfortunately, this is false.

Let $f(x) = 2$ and $g(x) = x^2$. By the Power Rule, we know that $\frac{d}{dx}[f(x)\,g(x)] = 4x$. However, if we were to treat the two as separate functions, under the assumption that $\frac{d}{dx}[f(x) \cdot g(x)] = f'(x) \cdot g'(x)$, we would find that $\frac{d}{dx}[f(x) \cdot g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$ which in this case is $0 \cdot 2x = 0$, which is evidently not correct.

So how do we calculate the derivative of the product of two functions?

We use the **PRODUCT RULE**

For a differentiable function $h(x)$ of the form $f(x)\,g(x)$, $\frac{d}{dx}h(x) = f'(x)\,g(x) + f(x)\,g'(x)$

To see why this is true, you can view the proof at the end of this unit.

How can we use the Product Rule?

Say we were asked the derivative of $x^2 ln(x)$. Using the Product Rule, we set $f(x) = x^2$ and $g(x) = ln(x)$ and solve:

$$\frac{d}{dx}x^2 ln(x) = 2x\,ln(x) + x^2\left(\frac{1}{x}\right) = 2x\,ln(x) + x.$$

It is crucial that you know the Product Rule , for like everything else in this unit, it will most definitely be tested on the AP Calculus ® exam.

<u>Exercise 2.3:</u>
Differentiate the following functions. You can check your answer at the end of this unit:

1. $\frac{d}{dx}x^7 ln(x)$
2. $\frac{d}{dx}6x\,sin(x)$
3. $\frac{d}{dx}sin(x)cos(x) + ln(x) + x^3e^x$
4. $\frac{d}{dx}2x^2 3^x$
5. $\frac{d}{dx}ln(x)e^x$

<u>The Quotient Rule:</u>

For differentiable functions $f(x)$ and $g(x)$, what is the derivative of $\frac{f(x)}{g(x)}$?

The **QUOTIENT RULE** gives us the answer!

For a differentiable function $h(x) = \frac{f(x)}{g(x)}$, $g(x) \neq 0$, $\frac{d}{dx} h(x) = \frac{f'(x)\,g(x) + f(x)\,g'(x)}{g^2(x)}$

How can we use the Quotient Rule?

Say we were asked the derivative of $\frac{x^3}{sin(x)}$. Using the Product Rule, we set $f(x) = x^3$ and $g(x) = sin(x)$ and solve:

$$\frac{d}{dx}\frac{x^3}{sin(x)} = \frac{3x^2 sin(x) - x^3 cos(x)}{sin^2(x)}$$

It is crucial that you know the Quotient Rule , for like everything else in this unit, it will most definitely be tested on the AP Calculus ® exam.

<u>Exercise 2.4:</u>

Differentiate the following functions. You can check your answer at the end of this unit.

1. $\frac{d}{dx}\frac{x^2+2x}{e^x}$
2. $\frac{d}{dx}\frac{1}{4sin(x)}$
3. $\frac{d}{dx}\frac{cos(x)}{sin(x)}$
4. $\frac{d}{dx}\frac{e^x}{x^2}$
5. $\frac{d}{dx}\frac{\sqrt{x}}{4ln(x)}$

<u>The Chain Rule:</u>

For differentiable functions $f(x)$ and $g(x)$, what is the derivative of $(f \circ g)(x)$? Or $(g \circ f)(x)$?

How can we find the derivative of a composite function?

For example, we know that $\frac{d}{dx} sin\,(x) = cos\,(x)$ and $\frac{d}{dx} x^2 = 2x$. But how can we compute the derivative of $sin\,(x^2)$?

Recall the definition of a composite function: For a function $h\,(x) = (f \circ g)\,(x)$, the function f is applied to the result of first applying the function g to x. In other words, was can think of g as the "inner function" whose output depends on the value of x, and f as the "outer function" whose value is dependent upon the value of g.

But how do we find the derivative of $(f \circ g)\,(x)$? We must use the **CHAIN RULE.**

If f and g are differentiable and $h(x) = (f \circ g)\,(x)$, $\frac{d}{dx} h\,(x) = f'\,(g\,(x)\,) \cdot g\,(x)$

In other words, to take the derivative of a composite function, we first differentiate the "outer" function while keeping the "inner" function unchanged, then multiply by the derivative of the "inner" function. If we have a function that is the composition of 3 or more functions, we use the same process: differentiate from the outermost function and multiply by the derivatives of the successive inner functions.

Say we had a function $y\,(x) = f\,(g\,(h\,(x)\,)\,)$.
To differentiate y, we would first differentiate f, keeping the input - $(f \circ g)\,(x)$ - unchanged. Like this:
$f'\,(g\,(h\,(x)\,)\,)$.

Then we differentiate the "next" outermost function, which in this case is $g\,(x)$, and multiply the resultant expression by $f'\,(g\,(h\,(x)\,)\,)$:

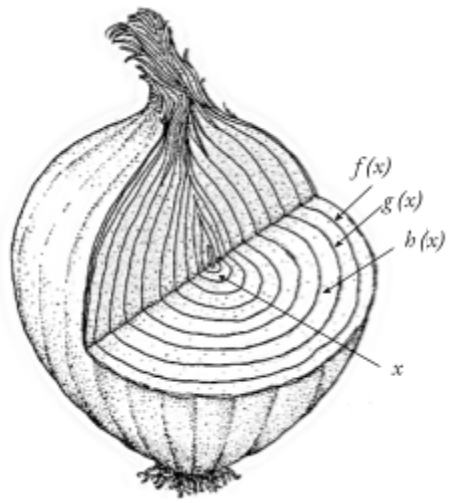
$f'\,(g\,(h\,(x)\,)\,) \cdot g'\,(h\,(x)\,)$

Finally we differentiate the innermost function, $h\,(x)$ as follows:

$f'\,(g\,(h\,(x)\,)\,) \cdot g'\,(h\,(x)\,) \cdot h'\,(x)$

And now we're done! The derivative of $y\,(x)$ is $f'\,(g\,(h\,(x)\,)\,) \cdot g'\,(h\,(x)\,) \cdot h'\,(x)$.

It's a bit like peeling an onion: We first "peel" (differentiate) the outermost "layer" (function), then we work our way in until we have "peeled" the entire "onion" (differentiated the entire function).



Let's work through a few practice problems:

1. Differentiate $sin\,(x^2)$.

As a composite function, we can dub $g\,(x) = x^2$ as the 'inner" function and $f\,(x) = sin\,(x)$ as the "outer" function. Using the Chain Rule, we first would differentiate the "outer function" (which in this case is $sin\,(x)$), then multiply that expression by the derivative of the "inner function" (which is x^2).

Thus, $\frac{d}{dx} sin\,(x^2) = cos\,(x^2) \cdot 2x$

2. Differentiate $tan\,(x^2) \cdot \frac{1}{e^x}$.

To compute this derivative, we must first use the Product Rule, since we are dealing with the product of 2 functions:

$$\frac{d}{dx} tan\,(x^2) \cdot \frac{1}{e^x} = \frac{d}{dx}\,[tan\,(x^2)] \cdot \frac{1}{e^x} + \frac{d}{dx}[\frac{1}{e^x}] \cdot tan\,(x^2) \;.$$

Now we can use the Chain Rule to differentiate:

$$\frac{d}{dx}\,[tan\,(x^2)] \cdot \frac{1}{e^x} + \frac{d}{dx}[\frac{1}{e^x}] \cdot tan\,(x^2) = \frac{2x\,sec^2\,(x^2)}{e^x} + (-\frac{1}{e^x}) tan\,(x^2)$$

So our final answer is $\frac{2x\,sec^2\,(x^2) - tan\,(x^2)}{e^x}$.

<u>Exercise 2.5: MIXED REVIEW</u>

Differentiate the following functions. You can check your answer at the end of this unit.

1. $\frac{d}{dx} sin^2 (x)$

2. $\frac{d}{dx} 3^x ln (\sqrt{x})$

3. $\frac{d}{dx} \frac{tan (x^2)}{5x^4 + 3x - 7}$

4. $\frac{d}{dx} e^{ln (x^4)}$

5. $\frac{d}{dx} x^{cot (2x)}$

6. Write the equation of the line tangent to $y = sin (x) + \frac{1}{x}$ at $x = \frac{\pi}{6}$

7. Write the equation of the line tangent to $f (x) = e^{2x}$ at $x = 2$

<u>Answer Key</u>

Exercise 2.2

1. $\frac{d}{dx}x^7 = 7x^6$

2. $\frac{d}{dx}3x^4 = 12x^3$

3. $\frac{d}{dx}x^5 + 3x^2 + 7x = 5x^4 + 6x + 7$

4. $\frac{d}{dx}\frac{1}{x} = \frac{d}{dx}x^{-1} = -x^{-2} = \frac{1}{x^2}$

5. $\frac{d}{dx}\ 5x^2 - \frac{1}{x^2} = 10x + \frac{2}{x^3}$

6. $\frac{d}{dx}\sqrt{x} = \frac{d}{dx}x^{\frac{1}{2}} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$

7. $\frac{d}{dx}\frac{1}{\sqrt[3]{x}} = \frac{d}{dx}x^{-\frac{1}{3}} = -\frac{1}{3}x^{-\frac{2}{3}} = -\frac{1}{3x^{\frac{2}{3}}}$

Exercise 2.3

1. $\frac{d}{dx}x^7 ln(x) = 7x^6 ln(x) + x^6$

2. $\frac{d}{dx}6x\ sin(x) = 6sin(x) + 6x\ cos(x)$

3. $\frac{d}{dx}sin(x)cos(x) + ln(x) + x^3e^x = cos^2(x) - sin^2(x) + \frac{1}{x} + e^x\ (x^3 + 3x^2)$

4. $\frac{d}{dx}2x^2 3^x = 4x\ (3^x) + 2x^2\ (3^x ln(3)\ )$

5. $\frac{d}{dx}ln\ (x)e^x = e^x(\frac{1}{x} + ln(x)\ )$

Exercise 2.4

1. $\frac{d}{dx}\frac{x^2+2x}{e^x} = \frac{(2x+2)e^x - (x^2+2x)e^x}{e^{2x}} = \frac{(-x^2+2)}{e^x}$

2. $\frac{d}{dx}\frac{1}{4sin(x)} = \frac{-4cos(x)}{16sin^2(x)} = \frac{-1}{4}csc(x)cot(x)$

3. $\frac{d}{dx}\frac{cos(x)}{sin\ (x)} = \frac{-sin^2(x) - cos^2(x)}{sin^2(x)} = \frac{-[sin^2(x) + cos^2(x)]}{sin^2(x)} = \frac{-1}{sin^2(x)} = -csc^2(x)$

4. $\frac{d}{dx}\frac{e^x}{x^2} = \frac{e^x(x^2+2x)}{x^4}$

5. $\frac{d}{dx}\frac{\sqrt{x}}{4ln\ (x)} = \frac{\frac{4\ ln\ (x)}{2\sqrt{x}} - \frac{4\sqrt{x}}{x}}{16\ ln^2(x)} = \frac{\frac{2\sqrt{x}\ ln\ (x) - 4\sqrt{x}}{x}}{16\ ln^2(x)} = \frac{\sqrt{x}\ ln\ (x) - 2\sqrt{x}}{8\ ln^2\ (x)x}$

Exercise 2.5: MIXED REVIEW

1. $\frac{d}{dx} \sin^2(x) = 2\sin(x)\cos(x) = \sin(2x)$

2. $\frac{d}{dx} 3^x \ln(\sqrt{x}) = 3^x \ln(3) \ln(\sqrt{x}) - \frac{3^x}{2x}$

3. $\frac{d}{dx} \frac{\tan(x^2)}{5x^4 + 3x - 7} = \frac{2x\sec^2(x^2)\cdot(5x^4+3x-7) - \tan(x^2)\cdot(20x^3+3)}{(5x^4+3x-7)^2}$

4. $\frac{d}{dx} e^{\ln(x^4)} = 4x^3$

5. $\frac{d}{dx} e^{\cot(2x)} = -2e^{\cot(2x)}\csc^2(2x)$

6. $y - (\frac{1}{2} + \frac{6}{\pi}) = (\frac{\sqrt{3}}{2} - \frac{36}{\pi^2})(x - \frac{\pi}{6})$

To write the equation of a line, we need 2 points (x_1, y_1) and the slope, m, of the line.

To find the slope at $x = \frac{\pi}{6}$, we first find the derivative of y, and evaluate at $x = \frac{\pi}{6}$

$\frac{d}{dx}\Big|_{x=\frac{\pi}{6}} \sin(x) + \frac{1}{x} = \cos(\frac{\pi}{6}) - \frac{1}{(\frac{\pi}{6})^2} = \frac{\sqrt{3}}{2} - \frac{36}{\pi^2}$

To determine the coordinates, we evaluate $\frac{\pi}{6}$ into the original function y:

$y(\frac{\pi}{6}) = \sin(\frac{\pi}{6}) + \frac{1}{(\frac{\pi}{6})} = \frac{1}{2} + \frac{6}{\pi}$

Using the coordinate $(\frac{\pi}{6}, \frac{1}{2} + \frac{6}{\pi})$ and the instantaneous slope at $x = \frac{\pi}{6}$, we can now write the equation of the line in point-slope form:

$y - (\frac{1}{2} + \frac{6}{\pi}) = (\frac{\sqrt{3}}{2} - \frac{36}{\pi^2})(x - \frac{\pi}{6})$.

7. $y - e^4 = 2e^4(x - 2)$

Bonus: Proof of the Product Rule

We wish to find $\frac{d}{dx} f(x)\, g(x)$, assuming that f and g are differentiable.

We start by using the definition of the derivative:

$$\lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}$$

Add, then subtract $f(x+h)g(x)$ in the numerator. Note that this is equivalent to the previous expression.

$$\lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h}$$

Factor:

$$\lim_{h\to 0} \frac{f(x+h)[g(x+h) - g(x)] + g(x)[f(x+h) - f(x)]}{h}$$

Rewrite:

$$\lim_{h\to 0} \frac{f(x+h)[g(x+h) - g(x)]}{h} + \lim_{h\to 0} \frac{g(x)[f(x+h) - f(x)]}{h}$$

$$\lim_{h\to 0} \frac{f(x+h)[g(x+h) - g(x)]}{h} + \lim_{h\to 0} \frac{g(x)[f(x+h) - f(x)]}{h}$$

Since f is differentiable, we know that f must also be continuous. Therefore, we know $\lim_{h\to 0} f(x+h) = f(x)$, which allows us to rewrite the above limit as follows:

$$f(x) \lim_{h\to 0} \frac{[g(x+h) - g(x)]}{h} + g(x) \lim_{h\to 0} \frac{f(x+h) - f(x)]}{h}$$

Using the definition of the derivative, we can now conclude our proof.

$$f(x) \lim_{h\to 0} \frac{[g(x+h) - g(x)]}{h} + g(x) \lim_{h\to 0} \frac{f(x+h) - f(x)]}{h} = f(x)g'(x) + f'(x)g(x).$$